

Linear Algebra—Winter/Spring 2026

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Assignment Problems

Questions in the assignment component of the end-semester exam will be taken only from this list of assignment problems.

1. Prove that the dimension of the vector space $\mathbb{R}$ over the field $\mathbb{Q}$ is infinite.
2. Let m, n be positive integers. Suppose R is an $m \times n$ row-reduced echelon matrix over some field F and denote its non-zero row vectors by $\rho_1, \rho_2 \dots \rho_r$. Let W be its row space and suppose the vector $\beta = (b_1, b_2 \dots b_n)$ is in W . Below, we use k_i to denote the index of the first non-zero column in the row vector ρ_i .
 - i. Find the coefficients $c_1, \dots, c_r$ (in terms of $b_1, \dots, b_n$) when β is written as $\beta = c_1\rho_1 + \dots c_r\rho_r$.
 - ii. Let $1 \leq s \leq r$. Prove that there is exactly one vector σ_s in W such that the k_s th coordinate of $\sigma_s = 1$ and the remaining k_i th coordinates (for $i \in \{1 \dots r\} \setminus s$) of σ_s are zero.
 - iii. Is $\sigma_s = \rho_s$? Prove your answer.
3. Let V and W be vector spaces over the field F and let T be a linear transformation from V to W . Suppose the vectors $\{\alpha_1 \dots \alpha_k\}$ form a basis for the null space of T . Let $\{\alpha_{k+1}, \dots, \alpha_n\}$ be vectors such that $\{\alpha_1, \dots, \alpha_n\}$ are a basis for V . Show that $\{T\alpha_{k+1}, \dots, T\alpha_n\}$ are linearly independent and span the range of T .
4. Prove that a function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation if and only if there exists a unique $m \times n$ matrix A such that $T(\alpha) = A\alpha$, for all $\alpha \in \mathbb{R}^n$. Let $L(\mathbb{R}^n, \mathbb{R}^m)$ denote the vector space of all linear transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$. Find a basis of $L(\mathbb{R}^n, \mathbb{R}^m)$.
5. Suppose $U_1, \dots, U_m$ are finite-dimensional subspaces of a vector space V such that $U_1 + \dots + U_m$ is a direct sum. Prove that $U_1 + \dots + U_m$ is finite-dimensional and
$$\dim(U_1 \oplus \dots \oplus U_m) = \dim U_1 + \dots + \dim U_m.$$
6. Let T be a linear operator.
 - a. Show that T has a 1-dimensional invariant subspace iff T has an eigenvalue.
 - b. Show that $\text{range}(T)$ is an invariant subspace of T .
 - c. Show that $\text{nullspace}(T)$ is an invariant subspace of T . Given that T has an eigenvalue λ , what can you say about $\text{nullspace}(T - \lambda\mathbb{I})$?

7. Let V be a vector space and let $S, T \in \mathcal{L}(V)$ be linear operators such that $ST = TS$. Prove that the null space of S and the range of S are invariant under T .
8. Suppose $T \in \mathcal{L}(V)$ and $v_1, \dots, v_n$ is a basis of V . Show that the following are equivalent.
 - a. The matrix of T with respect to $v_1, \dots, v_n$ is upper triangular.
 - b. $Tv_j \in \text{span}(v_1, \dots, v_j)$ for each $j \in \{1, \dots, n\}$.
 - c. $\text{span}(v_1, \dots, v_j)$ is invariant under T for each $j \in \{1, \dots, n\}$.
9. Let $T \in \mathcal{L}(\mathbb{F}^3)$ be the linear operator defined by

$$T(z_1, z_2, z_3) = (2z_2, 0, 5z_3).$$

- a. Find all eigenvalues of T .
 - b. For each eigenvalue, determine the corresponding eigenspace.
10. Let V be a finite-dimensional vector space over a field $\mathbb{F}$, and let $T \in \mathcal{L}(V)$. Let B be any basis of V , and let $[T]_B$ denote the matrix representation of T with respect to the basis B .
 - a. Prove that T is invertible if and only if the matrix $[T]_B$ is invertible.
 - b. Let $\lambda \in \mathbb{F}$. Prove that λ is an eigenvalue of T if and only if the matrix $[T - \lambda I]_B$ is not invertible.
11. Suppose the matrix of a linear operator T , relative to the standard basis, is given by

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & 4 \\ 0 & 0 & 12 \end{bmatrix}.$$

Can this matrix be diagonalised? If so, find the basis relative to which T is diagonal (showing how you obtained your result). If not, give a proof.

12. Suppose $T \in \mathcal{L}(V)$. Prove that if T has an upper-triangular matrix with respect to some basis $v_1, \dots, v_n$ of V , then T has an upper-triangular matrix with respect to the orthonormal vectors $e_1 \dots e_n$ obtained by the Gram Schmidt procedure, i.e. where $e_1 := v_1$ and for $1 < j \leq n$,

$$e_j := \frac{v_j - \langle v_j, e_1 \rangle e_1 - \dots - \langle v_j, e_{j-1} \rangle e_{j-1}}{c_j}$$

where c_j is a normalisation factor that ensures $\|e_j\|^2 = 1$.

13. On $\mathcal{P}_2(\mathbb{R})$, consider the inner product given by

$$\langle u, v \rangle = \int_0^1 p(x)q(x) \, dx.$$

Apply the Gram-Schmidt procedure to the basis $1, x, x^2$ to produce an orthonormal basis of $\mathcal{P}_2(\mathbb{R})$.

14. Fix $u \in V$ and $w' \in W$. Define $T \in \mathcal{L}(W, V)$ by

$$Tw = \langle w, w' \rangle u$$

for every $w \in W$. Find a formula for T^* .

15. Let V be a vector space over $\mathbb{C}$ and suppose $T \in \mathcal{L}(V)$ is a *normal* linear operator. Using Schur's Theorem, we know that the matrix of T is upper triangular on some orthonormal basis $e_1, \dots, e_n$, i.e.

$$\mathcal{M}(T, (e_1, \dots, e_n)) = \begin{bmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,n} \\ 0 & a_{2,2} & a_{2,3} & \dots & a_{2,n} \\ 0 & 0 & a_{3,3} & \ddots & \\ 0 & 0 & 0 & \dots & a_{n,n} \end{bmatrix}.$$

Suppose $a_{1,2} = 0$. Prove that $a_{2,3} = \dots = a_{2,n} = 0$.